

Math Guide

Arithmetic, randomness, and algebra for shielded payments

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Abstract

This volume of *The Zcash Arboretum* constructs the mathematical objects used by its cryptographic volumes. It assumes elementary integer arithmetic and the ability to follow an algorithm. We build ways to represent data, calculate with it and reason about random choices, introducing each required concept before applying it. Proofs establish the stated elementary results; facts imported without proof are identified explicitly. Knowing how a calculation works does not by itself establish that an attacker cannot reverse it.

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1 Language, integers, and encodings

A calculation on numbers and a message sent between computers are different objects. We first specify how messages represent numbers, which messages to reject, and whether different messages can denote the same number.

1.1 Objects, functions and assertions

A *set* is a collection whose members are distinct. We write $x \in S$ for membership, $S \subseteq T$ when every member of S belongs to T , and $\emptyset$ for the empty set. The symbol $S \cup T$ denotes the union: members in either set. The intersection $S \cap T$ contains members in both; $S \setminus T$ contains those in S but not T . The integer $|S|$ counts the elements of a finite set. An ordered pair (x, y) records both its entries and their order. The product $S \times T$ is the set of such pairs with $x \in S$ and $y \in T$. The notation S^n means ordered sequences, or tuples, of exactly n elements of S , including one empty sequence when $n = 0$.

A *function* $f : S \rightarrow T$ assigns exactly one output $f(x) \in T$ to each input $x \in S$. The sets S, T are its domain and codomain, respectively. Its image is $\text{im } f = \{f(x) : x \in S\}$. It is injective when equal outputs require equal inputs, surjective when $\text{im } f = T$, and bijective when both conditions hold. A bijection has an inverse function, which returns x from $f(x)$. The composition $g \circ f$ means first apply f , then g . A function that may reject has codomain $T \cup \{\perp\}$, where $\perp$ is a distinct failure symbol, not the number zero or an empty sequence.

An assertion has a truth value. The expressions $A \wedge B$, $A \vee B$, and $\neg A$ mean “both”, “at least one”, and “not”. The implication $A \Rightarrow B$ prohibits the case in which A is true and B false. Its contrapositive $\neg B \Rightarrow \neg A$ has exactly the same truth value. An equivalence $A \iff B$ asserts both implications. The quantifiers $\forall x \in S$ and $\exists x \in S$ mean “for every member” and “for at least one member”. Their order matters: $\forall x \exists y$ permits a different y for each x . An existence proof does not, by itself, give an efficient way to find the object it proves exists.

The natural numbers are $\mathbb{N} = \{0, 1, 2, \dots\}$; the integers $\mathbb{Z}$ include their negatives. We use induction as follows. To prove an assertion for every $n \in \mathbb{N}$, prove it at zero and prove that its truth at n implies its truth at $n + 1$. Equivalently, a nonempty set of natural numbers has a least member: a least counterexample could be neither zero nor the successor of an earlier non-counterexample. These elementary properties of integers are our starting mathematical assumptions, not consequences of cryptography.

1.2 Remainders and positional representations

Theorem 1.1 (Integer division). *For an integer a and positive integer b , unique integers q, r satisfy $a = qb + r$ and $0 \leq r < b$.*

Proof. Choose the least nonnegative integer in the set of integers $a - kb$. Such integers exist by taking k sufficiently negative. Its value r cannot be at least b , because $r - b$ would be a smaller nonnegative member. For uniqueness, two decompositions give $(q - q')b = r' - r$. The right side has absolute value less than b ; the only multiple of b with that property is zero. $\square$

We write $a \bmod b$ for this nonnegative remainder, including when a is negative. The floor $\lfloor x \rfloor$ is the greatest integer not exceeding x ; the ceiling $\lceil x \rceil$ is the least not smaller. For positive integer b , the quotient above is $\lfloor a/b \rfloor$.

Definition 1.2 (Digits and their value). For a base, or radix, $B \geq 2$, an n -digit little-endian se-

quence $(d_0, \dots, d_{n-1})$, with $0 \leq d_i < B$, represents

$$\text{val}_B(d_0, \dots, d_{n-1}) = \sum_{i=0}^{n-1} d_i B^i.$$

The sum sign adds one term for each index from zero through $n - 1$. The highest possible value is $B^n - 1$. Big-endian order reverses the digits, so its first digit has weight B^{n-1} .

Here $[0, B^n)$ denotes the integers from zero up to, but excluding, B^n . Repeated division by B constructs the digits of every integer in this interval: take the remainder as the next digit and continue with the quotient. Uniqueness of division proves uniqueness of every digit, starting with the least significant. Thus length- n digit strings are in bijection with the integers in that interval. A leading zero does not change the value, but it does change the length; lengths belong to the encoding specification.

A *bit* has radix two. The notation $\text{LE}_k(a)$ means the k -bit little-endian encoding of an integer $0 \leq a < 2^k$. A *byte* is an integer between zero and 255; a byte string is a sequence of bytes. We write $\text{LE}_n^{\text{byte}}(a)$ and $\text{BE}_n^{\text{byte}}(a)$ when an n -byte order must be explicit. A hexadecimal digit has radix 16; two such digits specify a byte. For instance $0x53$ denotes the integer $5 \cdot 16 + 3$, not a decimal string containing “53”. Concatenation $s \parallel t$ appends the sequence t to s ; the empty sequence is ϵ .

1.3 Canonical encodings and unambiguous tuples

Definition 1.3 (Canonical encoding). An encoding of a set S is an injective function $\text{enc} : S \rightarrow \{0, \dots, 255\}^*$, together with a decoder that returns x exactly on $\text{enc}(x)$ and returns $\perp$ on every other string. The star denotes all finite lengths, not just one fixed length.

Suppose an integer is required to lie in $[0, p)$ and occupies 32 bytes. The decoder must first recover the integer and check that bound. Reducing an arbitrary 32-byte integer modulo p defines a different operation: it accepts multiple encodings of the same remainder. Rejection and reduction cannot be interchanged merely because subsequent arithmetic produces the same value.

Concatenation of variable-length strings is not an injective tuple encoding: (a, bc) and (ab, c) produce the same concatenation. Fixed component lengths resolve the ambiguity. Alternatively, encode each component’s length in a specified format before its bytes. The format must itself be unambiguous and enforce its bounds. A type or domain tag distinguishes different constructions; it does not recover component boundaries that were never encoded.

Bitwise exclusive-or, written $\oplus$, adds corresponding bits modulo two. It is not ordinary integer addition: carries are discarded. A word is a fixed-length sequence of bits. For a w -bit word, rotation by j positions moves each bit j places, wrapping around the ends. A logical shift inserts zero bits and discards bits that pass the boundary; a rotation does neither. These operations, together with addition modulo 2^w , specify the integer operations used later by byte-oriented algorithms.

2 Prime fields and modular arithmetic

Finite arithmetic lets a protocol store every number in a bounded space. It also changes the meaning of equality. We first construct the arithmetic, then prove the bounds needed to recover statements about ordinary integers from statements about residues.

2.1 Divisibility, greatest common divisors and inversion

An integer d divides a , written $d \mid a$, if $a = dk$ for some integer k . A common divisor of a, b divides every integer combination $ua + vb$. For integers not both zero, $\text{gcd}(a, b)$ is their greatest positive common divisor.

Theorem 2.1 (Euclid and Bézout). *The repeated replacement $(a, b) \leftarrow (b, a \bmod b)$, starting with nonnegative integers and $b > 0$, terminates. Its last nonzero remainder is $g = \gcd(a, b)$. Integers u, v can be computed with $ua + vb = g$.*

Proof. Writing $a = qb + r$ shows that the common divisors of a, b are exactly those of b, r . Positive second entries strictly decrease, so a zero remainder is reached. At that point the positive first entry is the greatest common divisor. Initially, the two entries have coefficient pairs $(1, 0)$ and $(0, 1)$ as combinations of the original inputs. If the current coefficient pairs are (u, v) and (u', v') , the new remainder has pair $(u - qu', v - qv')$. Updating these pairs alongside the remainders gives the claimed identity constructively. $\square$

A positive integer $p > 1$ is prime when its only positive divisors are 1 and itself. If $p \nmid a$, then $\gcd(a, p) = 1$, so Theorem 2.1 gives $ua + vp = 1$. Consequently, if $p \mid ab$ and $p \nmid a$, multiplying the identity by b shows $p \mid b$. This is the prime-divisor property used below. Repeatedly factoring a composite integer proves existence of a product of primes, because each factor is smaller. Applying the prime-divisor property to two products and cancelling matching factors proves uniqueness, apart from ordering. This also justifies describing a positive integer by its prime-power factors.

2.2 Constructing a field

We write $a \equiv b \pmod{m}$ when $m \mid (a - b)$. Congruence is an equivalence relation: $a \equiv a$; $a \equiv b$ implies $b \equiv a$; and $a \equiv b, b \equiv c$ imply $a \equiv c$, all with the same modulus. These three properties are called reflexivity, symmetry and transitivity, respectively. Replacing either operand by a congruent integer leaves its sum and product congruent to the original result. We may therefore compute with representatives $0, \dots, m - 1$, reducing after each operation. Addition, multiplication and subtraction obey the usual associative, commutative and distributive laws because their integer counterparts do.

Definition 2.2 (Field). A field is a set with distinct elements $0, 1$, associative and commutative addition and multiplication, additive identity zero and multiplicative identity one, an additive inverse for every element, and a multiplicative inverse for every nonzero element. Multiplication distributes over addition.

Proposition 2.3 (Prime fields). *For prime p , the residues with arithmetic modulo p form a field $\mathbb{F}_p$. Its inverses are computed by extended Euclid. For composite $m > 1$, the residues modulo m do not form a field.*

Proof. All laws except inversion have just been inherited from the integers. For $0 < a < p$, Bézout gives $ua + vp = 1$, so $u \bmod p$ is an inverse of a . If $m = ab$ with $1 < a, b < m$, the two nonzero residues a, b have product zero. An inverse of a would imply $b = 0$, a contradiction. $\square$

In every field, $ab = 0$ implies $a = 0$ or $b = 0$: multiply by a^{-1} when $a \neq 0$. We may cancel a common nonzero factor for the same reason. Division a/b means ab^{-1} and is undefined when $b = 0$. An algorithm that uses a division must prove the denominator nonzero or specify a failure case.

Example 2.4 (A computed inverse). The Euclidean chain is

$$97 = 5 \cdot 17 + 12, \quad 17 = 12 + 5, \quad 12 = 2 \cdot 5 + 2, \quad 5 = 2 \cdot 2 + 1.$$

Back substitution gives $1 = 40 \cdot 17 - 7 \cdot 97$. Thus the inverse of 17 in $\mathbb{F}_{97}$ is 40. The example checker verifies the final identity independently of the displayed chain.

2.3 Efficient powers and simultaneous inverses

To compute a^e for a nonnegative integer e , initialise $r = 1, b = a, k = e$. While $k > 0$, multiply r by b if k is odd, replace b by b^2 , and replace k by $\lfloor k/2 \rfloor$. The invariant $rb^k = a^e$, interpreted before each iteration, proves the result. At most twice the bit length of e multiplications are needed. A secret-dependent branch in this description is mathematical pseudocode, not permission for a timing-dependent implementation.

There is also a useful way to invert many nonzero elements with one inversion. For $a_1, \dots, a_n \neq 0$, form prefix products $P_0 = 1, P_i = P_{i-1}a_i$. Invert P_n , obtaining $u = P_n^{-1}$. In reverse order, output $a_i^{-1} = uP_{i-1}$ and update $u \leftarrow ua_i$. The update makes $u = P_{i-1}^{-1}$, proving each output. Zero inputs must be removed and handled separately; the product method does not invert zero.

2.4 When a field equality is an integer equality

The notation $[a]_p$ denotes the residue of an integer a . The map $a \mapsto [a]_p$ preserves sums and products, but it is not injective on all integers. In particular, there is no order relation on $\mathbb{F}_p$ compatible with the usual ordered-field laws. The representative 96 is larger than 1 as an integer, but in $\mathbb{F}_{97}$ it is also -1 .

Lemma 2.5 (No wraparound). *If integers a, b satisfy $[a]_p = [b]_p$ and $|a - b| < p$, then $a = b$. More generally, reduction modulo p is injective on any interval of integers whose largest and smallest members differ by less than p .*

Proof. The difference is a multiple of p . Every nonzero such multiple has absolute value at least p . $\square$

Corollary 2.6 (A bounded balance equation). *Let $0 \leq u_i < U_i, 0 \leq v_j < V_j$, and let b be a signed integer with $|b| \leq B$. If*

$$\sum_i (U_i - 1) + \sum_j (V_j - 1) + B < p,$$

then the field equation $\sum_i [u_i]_p - \sum_j [v_j]_p = [b]_p$ implies the corresponding integer equation.

Proof. The absolute value of the difference between the two sides is bounded by the displayed sum. Apply Lemma 2.5. Tighter one-sided bounds are possible; this sufficient bound does not assume that either side is nonnegative. $\square$

A field constraint $b(b - 1) = 0$ forces $b \in \{0, 1\}$. Therefore constraints of this form together with $x = \sum_{i=0}^{k-1} 2^i b_i$ describe a k -bit integer in the field. When $2^k \leq p$, the represented integer is unique. When $2^k > p$, bit constraints alone do not prove that the representation is canonical modulo p ; an additional less-than- p condition is necessary. A comparison can be checked by a binary subtraction with explicit borrow bits, or by a range-constrained difference, provided its own field-to-integer bound is established.

Example 2.7 (A false integer balance). The equality $80 + 80 = 63$ holds in $\mathbb{F}_{97}$, but not in $\mathbb{Z}$. All three numbers are themselves canonical field elements. Canonical inputs do not prevent their sum from wrapping. A proof of a monetary balance must bound the complete signed difference, not merely each encoded input.

A field homomorphism is a function between fields preserving one, addition and multiplication. It necessarily preserves zero, negatives and inverses: apply it to $0 + 0 = 0$, $a + (-a) = 0$ and $aa^{-1} = 1$. It is injective, since a nonzero element mapped to zero would have its inverse product mapped to zero instead of one. Such an inclusion is also called a field embedding. A bijective field homomorphism is a field isomorphism; its inverse preserves the same operations by applying the original map and

using injectivity. These names concern preservation of both operations, not merely a bijection of the underlying sets.

For distinct primes p, r , conversion from a field element to a scalar means a specified sequence of operations: take its canonical integer representative, optionally check an additional bound, then reduce or canonically encode for the destination. This conversion is not a field homomorphism. Indeed, a map preserving one and addition would send $p \cdot 1 = 0$ in $\mathbb{F}_p$ to $p \cdot 1 \neq 0$ in $\mathbb{F}_r$. Coordinates and scalars must consequently retain their types even when their moduli have nearly the same size.

3 Groups and scalar multiplication

Field elements support two operations. For the objects called points we need only one operation and its inverse. The group laws organise that operation; they do not say that recovering a multiplier from its result is difficult.

3.1 The operation and its order

Definition 3.1 (Abelian group). An abelian group $(\mathbb{G}, +)$ has an associative and commutative operation, an identity $\mathcal{O}$ satisfying $P + \mathcal{O} = P$, and for each P an inverse $-P$ with $P + (-P) = \mathcal{O}$. Its order is its number of elements when finite. Subtraction means adding an inverse.

The identity and inverses are unique. If both I, J are identities, then $I = I + J = J$; if both Q, R invert P , then $Q = Q + (P + R) = (Q + P) + R = R$. Cancellation follows by adding the inverse to both sides of an equality. A subgroup is a subset containing the identity and closed under addition and inverses; the group laws restrict to it.

For an integer $k \geq 0$, define $[k]P$ to be the sum of k copies of P , with $[0]P = \mathcal{O}$. For negative k , set $[k]P = -[-k]P$. Associativity and induction give

$$[a + b]P = [a]P + [b]P, \quad [ab]P = [a]([b]P), \quad [a](P + Q) = [a]P + [a]Q.$$

The last identity uses commutativity. Double-and-add computes $[k]P$ by the same binary method as field exponentiation, replacing multiplication by point addition and squaring by doubling. In a finite group, some positive multiple of P is the identity: among more multiples than group elements, two coincide. The least such positive integer is the order $\text{ord}(P)$.

Proposition 3.2 (Scalar reduction). *If P has order r , then $[a]P = [b]P$ exactly when $a \equiv b \pmod{r}$. The subgroup generated by P has exactly r elements.*

Proof. Divide $a - b$ by r . Its remainder s satisfies $[a - b]P = [s]P$. An identity with $0 < s < r$ would contradict minimality of r . Conversely, a multiple of r acts as zero. The multiples indexed by $0, \dots, r-1$ are therefore distinct and include all multiples. $\square$

3.2 Counting subgroups and prime-order groups

For a subgroup H , a coset is the set $P + H = \{P + Q : Q \in H\}$. Translation is a bijection from H onto this set. Two cosets that share an element are equal: from $P + h = Q + h'$ one obtains $P - Q = h' - h \in H$, which translates one description into the other. Cosets thus partition a finite group into equally sized pieces.

Theorem 3.3 (Lagrange's counting theorem). *The order of a subgroup divides the order of a finite group. In particular, the order of each element divides the group order.*

Proof. Count the partition into cosets, then apply the result to the subgroup generated by the element. $\square$

If the group order is prime r , every nonidentity point has order r ; hence every such point generates the whole group. A group generated by one element is called cyclic. After choosing a generator G , the function $a \mapsto [a]G$ is a bijection from $\mathbb{F}_r$ to the group. Multiplication by a nonzero scalar is also a bijection, with inverse multiplication by its field inverse. These are algebraic statements even when computing the inverse of the first bijection from the encoded point is not known to be efficient.

The nonzero elements $\mathbb{F}_p^\times$ form a multiplicative group of order $p - 1$. Lagrange therefore proves Fermat's identity $a^{p-1} = 1$ for $a \neq 0$. The formula $a^{-1} = a^{p-2}$ is an alternative inversion algorithm. Unlike $\mathbb{F}_p^\times$, a prime-order point group has no nontrivial proper subgroups. When a larger point group contains a prime-order subgroup of order r , the ratio of the group order to r is the cofactor. Multiplying by a cofactor and checking membership are different operations; a protocol must specify which it requires.

3.3 Linear combinations of points

A map $f : \mathbb{G} \rightarrow \mathbb{G}'$ preserving addition is a homomorphism. It preserves the identity, inverses and integer scalar multiplication by the laws already proved. A bijective homomorphism is an isomorphism; it preserves the abstract operation, not necessarily the representation or the cost of computing its inverse.

For points $G_0, \dots, G_{n-1}$ in a prime-order group and scalars $a_i \in \mathbb{F}_r$, the multiscalar multiplication is

$$\langle \mathbf{a}, \mathbf{G} \rangle = \sum_{i=0}^{n-1} [a_i]G_i.$$

It is linear in each argument where addition and scalar multiplication are defined. Evaluating each term by double-and-add is a complete algorithm; faster implementations share work among the terms but do not change the result. Relative to any fixed generator, every point has a scalar representation. Thus a list of group generators always satisfies some relation $\sum_i [a_i]G_i = \mathcal{O}$ with coefficients not all zero when at least two bases are present. Cryptographic uses of several generators rely instead on a claim that certain relations between their scalars are computationally unknown.

4 Probability and computation

A random choice is specified by a distribution, not by the word “random”. In particular, a value can be uniform before an observation and strongly biased after it. We develop the counting and conditioning rules needed to make such distinctions explicit.

4.1 Real quantities and approximation

The rational numbers $\mathbb{Q}$ are fractions a/b of integers, with $b \neq 0$, identifying a/b and c/d when $ad = bc$. Some limits of increasingly accurate rational approximations, such as the positive square root of two, are not rational. We use the real numbers $\mathbb{R}$ to include such limits. Concretely, an exact real can be specified by nested closed intervals with rational endpoints and lengths tending to zero. The intervals describe approximations at arbitrary accuracy, not a promise that every real has a finite decimal expansion.

The foundational property used here is completeness: every nonempty set of reals bounded above has a least upper bound, its supremum. This is an explicit property of the real number system, alongside its ordered-field laws; we do not require the reader to know calculus. The infimum is the corresponding greatest lower bound. Nested nonempty closed bounded intervals whose lengths tend to

zero have exactly one common point: take the supremum of their left endpoints, and use the shrinking lengths for uniqueness. This explains the interval representation and the convergence of binary search.

The notation $[a, b]$ includes both endpoints; (a, b) excludes both, with mixed brackets for one included endpoint. A sequence x_n tends to x if, for every accuracy $\epsilon > 0$, eventually $|x_n - x| < \epsilon$. A function is continuous at x if arbitrarily small changes in its input near x make its output arbitrarily close to $f(x)$: for every $\epsilon > 0$ there is $\delta > 0$ such that $|u - x| < \delta$ implies $|f(u) - f(x)| < \epsilon$. Finite sums and products preserve limits, as follows by bounding the difference of sums or expanding a difference of products. Quotients preserve them when the limiting denominator is nonzero. These elementary rules justify the approximations used below.

4.2 Experiments, conditioning and expectation

A finite probability space consists of a set Ω of outcomes and nonnegative weights $w(\omega)$ summing to one. An event is a subset $A \subseteq \Omega$, with probability $\Pr[A] = \sum_{\omega \in A} w(\omega)$. In a uniform space all weights equal $1/|\Omega|$; this definition requires $\Omega \neq \emptyset$. The notation $X \leftarrow S$ means choose uniformly from a nonempty finite set S , using fresh independent randomness unless another distribution is specified. For a finite distribution, its support is the set of outcomes with positive probability.

For $\Pr[B] > 0$, conditional probability is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

Events are independent when $\Pr[A \cap B] = \Pr[A] \Pr[B]$. For mutually independent trials this factorisation extends to every selection of events from the different trials; pairwise independence alone does not establish it. A partition B_i of the outcome space gives the total-probability identity $\Pr[A] = \sum_i \Pr[B_i] \Pr[A \mid B_i]$, omitting zero-probability parts. Rearranging the definition gives Bayes' identity $\Pr[B \mid A] = \Pr[A \mid B] \Pr[B] / \Pr[A]$.

A random variable is a function on outcomes. For a real-valued finite random variable, its expectation is the weighted average $\mathbb{E}[X] = \sum_{\omega} w(\omega) X(\omega)$. Rearranging finite sums proves linearity $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, without an independence requirement. An indicator 1_A is one on A and zero elsewhere; its expectation is $\Pr[A]$. The variance $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$ measures squared deviation from the mean. Expanding the square gives $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$.

Lemma 4.1 (Basic probability bounds). *For events $A_1, \dots, A_n$, $\Pr[\bigcup_i A_i] \leq \sum_i \Pr[A_i]$. For nonnegative X and $t > 0$, $\Pr[X \geq t] \leq \mathbb{E}[X]/t$. Consequently, $\Pr[|X - \mathbb{E}X| \geq t] \leq \text{Var}(X)/t^2$.*

Proof. Pointwise, $1_{\bigcup_i A_i} \leq \sum_i 1_{A_i}$ and $X \geq t 1_{X \geq t}$. Taking expectations gives the first two bounds. Apply the second to $(X - \mathbb{E}X)^2$ for the third. $\square$

These are the union, Markov and Chebyshev bounds, respectively. None permits replacing a conditional distribution by an unconditional one. For instance, if a hidden integer is uniform on a set of size N , then after one incorrect guess is revealed, the next distinct guess hits with probability $1/(N-1)$, not $1/N$.

4.3 Sampling, collisions and concentrated counts

For q independent uniform choices from N objects, a fixed pair collides with probability $1/N$. The union bound over pairs gives probability at most $q(q-1)/(2N)$ for any collision. For $q \leq N$, multiplying the probabilities of avoiding all earlier choices gives the exact no-collision probability

$$\prod_{j=0}^{q-1} \left(1 - \frac{j}{N}\right).$$

The pair count is useful even when it exceeds one, but then its upper bound is uninformative and should be capped at one.

If n independent trials succeed with probability θ , the count X has binomial distribution

$$\Pr[X = k] = \binom{n}{k} \theta^k (1 - \theta)^{n-k}, \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $n! = 1 \cdot 2 \cdots n$, with $0! = 1$. The coefficient counts the choices of the k successful positions. Summing over all subsets proves that the probabilities sum to one. Writing the count as a sum of indicators gives $\mathbb{E}X = n\theta$; independence makes the cross terms in its variance vanish, giving $\text{Var}(X) = n\theta(1 - \theta)$.

Without replacement, sampling n distinct objects uniformly from N objects, of which K are marked, instead gives

$$\Pr[X = k] = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

Impossible choices have coefficient zero. Count the marked and unmarked subsets to obtain this formula. These two distributions cannot be exchanged simply because the sample size is small relative to the population; an approximation needs its own error bound.

For sharper upper tails, apply Markov to t^X , with $t > 1$. Let X be a sum of independent zero/one variables, where variable i equals one with probability θ_i . For an integer threshold a , they satisfy the finite bound

$$\Pr[X \geq a] \leq t^{-a} \prod_i (1 - \theta_i + \theta_i t). \quad (1)$$

Indeed, the expectation of a product of independent variables factors, and $\mathbb{E}[t^{1_A}] = 1 - \theta + \theta t$. In the equal-probability case, for $0 < \theta < 1$ and $n\theta < a < n$, substituting $t = a(1 - \theta)/(\theta(n - a)) > 1$ yields

$$\Pr[X \geq a] \leq \left(\frac{n\theta}{a}\right)^a \left(\frac{n(1 - \theta)}{n - a}\right)^{n-a}.$$

This explicit bound needs no replacement distribution and no unstated limit. If $\theta = 0$, then $X = 0$ certainly and every positive threshold has probability zero; if $\theta = 1$, then $X = n$ certainly. Related sampling models must justify their own expectation factorisation or another bound replacing it.

4.4 Countable experiments and waiting times

An algorithm may retry indefinitely. A finite probability space no longer describes all its possible running times. For a countable set of outcomes, meaning outcomes that can be listed by natural-number indices, we assign nonnegative weights with sum one; an infinite sum of nonnegative terms is the supremum of its finite partial sums. This definition permits rearranging such terms, since both arrangements have the same finite subsums. Probabilities of increasing events tend to the probability of their union by the same reasoning. We need no general integration theory for these uses.

For an integer-valued $T \geq 0$, write $T = \sum_{j \geq 0} 1_{T > j}$ pointwise. Nonnegative rearrangement gives the tail-sum formula

$$\mathbb{E}T = \sum_{j \geq 0} \Pr[T > j], \quad (2)$$

allowing an infinite expectation. With independent success probability $\theta > 0$ on every trial, the number $T \geq 1$ of trials up to and including the first success has $\Pr[T > j] = (1 - \theta)^j$ and $\mathbb{E}T = 1/\theta$. The geometric series identity follows by summing its first m terms and taking the limit. Success occurs eventually with probability one, but that statement by itself would not imply finite expected time.

Waiting for $r \geq 1$ successes requires a number of failures K with

$$\Pr[K = k] = \binom{k + r - 1}{k} \theta^r (1 - \theta)^k.$$

The final trial must succeed; choose the k failures among the preceding $k+r-1$ positions. Equivalently, the total waiting time is a sum of r geometric waiting times, so it has expectation r/θ . This is the negative-binomial waiting model. It differs from fixing an elapsed time first and counting successes during it.

4.5 Integrals and continuous distributions

For a bounded function f on $[a, b]$, divide the interval at finitely many points $a = x_0 < \dots < x_n = b$. Multiplying each subinterval's width by the infimum of f there gives a lower sum; using its supremum gives an upper sum. Refining a partition cannot decrease the lower sum or increase the upper sum. If the supremum of all lower sums equals the infimum of all upper sums, their common value is the integral $\int_a^b f(x) dx$. It is thus a quantity constructed by progressively tighter finite arithmetic bounds. Choosing one point in each subinterval instead gives a Riemann sum, the sum of its function value times that subinterval's width. It lies between the lower and upper sums. The mesh is the largest width in the partition.

Continuous functions on a closed bounded interval have such an integral. Here are the needed details. Repeated interval bisection shows that any sequence in $[a, b]$ has a subsequence converging to a point in that interval: retain a half containing infinitely many sequence entries at each stage and choose entries with increasing indices. If a continuous function were unbounded, a sequence on which its absolute value grows beyond every bound would have such a subsequence, contradicting continuity at its limiting point. Likewise, if there were no uniform input tolerance for a fixed output accuracy, there would be pairs u_n, v_n with $|u_n - v_n| \rightarrow 0$ but a fixed positive separation of their outputs. A converging subsequence of u_n , and hence of v_n , contradicts continuity. Thus for any $\epsilon > 0$ we can choose a mesh small enough that the upper and lower values in each subinterval differ by less than ϵ . The upper and lower sums differ by at most $\epsilon(b-a)$, proving integrability.

The finite sums prove positivity, additivity over adjoining intervals, and linearity in the integrand. They also prove $|\int_a^b f| \leq (b-a) \sup |f|$. Define an integral with reversed endpoints by $\int_b^a f = -\int_a^b f$. A shift and rescaling of the variable, $t = c + du$, rescales every partition width by $|d|$; accounting for endpoint order gives

$$\int_{\alpha}^{\beta} f(c + du) du = \int_{c+d\alpha}^{c+d\beta} f(t) dt.$$

This substitution identity is proved directly from the partition sums, without assuming a differentiation rule.

A continuous probability density on an interval is a nonnegative function f whose integral over its domain is one. An interval probability is its integral on that interval. For an unbounded domain, define the total integral as the increasing limit over bounded intervals. A cumulative distribution function, abbreviated CDF, records $F(x) = \Pr[X \leq x]$. For a density on $[a, b]$, it is $F(x) = \int_a^x f(t) dt$ inside the interval, zero below it and one above it. Individual points have probability zero; probability is spread over intervals, not assigned a positive weight to each real point. For a density f , the expectation of a function $g(X)$ is $\int g(x)f(x) dx$ when the required integrals exist and the positive and negative parts are not both infinite. Our applications use continuous or piecewise-continuous integrands and their stated nonnegative limits. Piecewise-continuous means continuous on finitely many adjoining pieces of each bounded integration interval, with finite one-sided endpoint limits; the integral construction does not make every arbitrary bounded function integrable. For nonnegative quantities, rearrangement of nonnegative partition sums gives the continuous counterpart of (2):

$$\mathbb{E}X = \int_0^{\infty} \Pr[X > t] dt \quad (X \geq 0).$$

Indeed, approximate each value x from below by a sum of equally spaced threshold indicators, apply linearity to the finite sum, and take increasingly fine meshes and increasing cutoffs. Both sides are limits of the same nonnegative finite sums.

An ideal uniform real U on $[0, 1]$ can be represented by an infinite sequence of independent fair bits: $U = \sum_{i \geq 1} b_i 2^{-i}$. Fixing the first k bits selects an interval of length 2^{-k} with that probability. Approximating arbitrary interval endpoints by such binary intervals gives probability equal to interval length. Endpoints with two binary expansions have probability zero and do not change the distribution. This is an ideal continuous experiment, not an algorithm that can return infinitely many bits in finite time.

Lemma 4.2 (Sampling by the inverse CDF). *If F is continuous and strictly increasing from zero to one on an interval, then $X = F^{-1}(U)$ has CDF F , where U is uniform on $[0, 1]$. Endpoints or infinite endpoints can be defined by limits.*

Proof. Monotonicity gives $F^{-1}(U) \leq x$ exactly when $U \leq F(x)$. The latter event has probability $F(x)$. Existence of the inverse follows by binary search and continuity, and uniqueness follows by strict monotonicity. $\square$

A finite implementation may replace U by a uniform grid point $j/2^k$, $0 \leq j < 2^k$. For every threshold, the resulting interval count differs from its uniform-real probability by at most 2^{-k} . This is a bound on CDF error, not closeness for every possible event. The event that the output lies among the finitely many values the discrete sampler can return has probability one in the discrete experiment and zero in the continuous experiment. A protocol requiring a discrete distribution must specify its actual rounding or rejection algorithm rather than claiming that floating-point inversion is exact continuous sampling.

4.6 Natural logarithms, exponentials and their identities

For $x > 0$, define the natural logarithm constructively by

$$\ln x = \int_1^x \frac{dt}{t}.$$

The integrand is continuous on every positive closed bounded interval, so the preceding construction applies. Positivity of $1/t$ makes $\ln$ strictly increasing. The integral bound on a short interval proves its continuity. Splitting the interval at $a > 0$ and substituting $t = au$ proves

$$\ln(ab) = \ln a + \ln b, \quad \ln(1/a) = -\ln a, \quad \ln(a/b) = \ln a - \ln b.$$

For example, $\ln 2 \geq 1/2 > 0$, so $\ln(2^n) = n \ln 2$ grows without bound and $\ln(2^{-n}) = -n \ln 2$ decreases without bound. Continuity and binary search therefore give, for every real y , a unique positive real x with $\ln x = y$. Define $\exp(y)$ to be that real, and write $e = \exp(1)$. The product law above gives

$$\exp(u + v) = \exp(u) \exp(v), \quad \exp(-u) = 1/\exp(u).$$

For $a > 0$ and real u , define $a^u = \exp(u \ln a)$. This agrees with integer powers and supplies the exact meaning of fractional powers in sampling formulas. Logarithm to base $a \neq 1$ is $\log_a x = \ln x / \ln a$, the inverse of $u \mapsto a^u$.

The exponential also has the convergent series

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}. \quad (3)$$

To justify the identity without presupposing calculus, call the series $E(x)$. Absolute convergence follows because successive absolute terms eventually have ratio at most one half. Multiplying two absolutely convergent series and grouping by total degree, then using the finite binomial identity, gives $E(x+y) = E(x)E(y)$. These regroupings are legitimate: absolute tail sums bound the discarded terms.

In particular $E(x)E(-x) = 1$. The series is positive for $x \geq 0$, so it is positive everywhere. For $|h| \leq 1$, its terms of degree at least two have total absolute value at most a fixed constant times h^2 . Hence $(E(h) - 1)/h \rightarrow 1$, and the product identity makes E continuous everywhere.

The continuous function $L(x) = \ln E(x)$ is additive. Every continuous additive function equals cx for a fixed constant c : prove it first at integers, then rational fractions, then pass to real limits. For $v \rightarrow 0$, the integral bounds for $1/t$ between 1 and $1 + v$ give $\ln(1 + v)/v \rightarrow 1$. Substituting $v = E(h) - 1$ therefore gives $L(h)/h \rightarrow 1$, forcing $c = 1$. Thus $\ln E(x) = x$, which proves (3) from the inverse definition.

The shift and rescaling $t = 1 - x$ now yield the exact integral identity needed for inverse-distance sampling:

$$\int_a^b \frac{dx}{1-x} = \ln(1-a) - \ln(1-b), \quad a < b < 1. \quad (4)$$

The equivalent notation $\int dx/(1-x) = -\ln(1-x) + C$ means that differences of the expression at two endpoints give the definite integral; the arbitrary constant cancels. No knowledge of integration notation is assumed before this construction.

An exponential waiting time of rate $\lambda > 0$ has CDF $F(t) = 1 - \exp(-\lambda t)$ for $t \geq 0$, and zero for $t < 0$. It is sampled by $-\ln(1-U)/\lambda$; the event $U = 1$ has probability zero and is handled by a limiting endpoint. Its survival probability is $\exp(-\lambda t)$, so

$$\Pr[T > s + t \mid T > s] = \exp(-\lambda t) = \Pr[T > t].$$

This is the memoryless property, derived from the exponential product identity, not assumed from the phrase “waiting time”. Its density is $\lambda \exp(-\lambda t)$. One can verify the corresponding integral directly: the series gives $\exp(-\lambda h) = 1 - \lambda h + R(h)$, where $|R(h)| \leq Ch^2$ for a fixed constant on bounded intervals. Summing the increments of $1 - \exp(-\lambda t)$ along a partition differs from the density’s Riemann sum by at most a constant times the largest partition width, which tends to zero. The tail-integral formula and this density calculation give $\mathbb{E}T = 1/\lambda$.

For completeness, a Poisson distribution with parameter $\lambda > 0$ assigns probability $e^{-\lambda} \lambda^k / k!$ to $k \geq 0$. One may define its normalising constant directly by $e^\lambda = \sum_{k \geq 0} \lambda^k / k!$; the series converges because the ratio of consecutive terms eventually becomes less than one half. Reindexing nonnegative sums gives mean and variance λ . A counting process records the number $N(t)$ of events up to each time $t \geq 0$, starting from zero. An increment $N(t+h) - N(t)$ counts events in the subsequent interval. Describing this as a Poisson process additionally assumes that increments on disjoint intervals are independent and have this distribution with λ proportional to the interval length. That is a modelling hypothesis about event arrivals, not a consequence of the binomial formula or of a hash function’s definition.

4.7 Statistical distance and coupled executions

Definition 4.3 (Statistical distance). For distributions P, Q on the same finite or countable set,

$$\text{SD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

Let $D = \{x : P(x) > Q(x)\}$. Since both distributions sum to one, the total positive excess equals the total negative deficit, so $\text{SD}(P, Q) = P(D) - Q(D)$. For any event A , the difference $P(A) - Q(A)$ cannot exceed that positive excess; reversing the roles gives $|P(A) - Q(A)| \leq \text{SD}(P, Q)$. Thus distance is exactly the largest difference in probability of an observable event.

The triangle inequality for absolute values proves $\text{SD}(P, R) \leq \text{SD}(P, Q) + \text{SD}(Q, R)$. Applying the same deterministic or randomised computation to both inputs cannot increase their distance: for each output event, its input-dependent probability is between zero and one, and summing against $P - Q$ is bounded by the total positive excess. This is the data-processing property.

Lemma 4.4 (Identical until a bad event). *Suppose two executions are realised on one probability space and their observable outputs agree except on an event B . The statistical distance between their output distributions is at most $\Pr[B]$.*

Proof. For any output event A , its two indicators differ only when the outputs differ, and their absolute difference is at most one. Their expectations therefore differ by at most $\Pr[B]$. Use the event characterisation of distance. $\square$

A shared realisation of two distributions is a coupling. Its usefulness is that it can preserve a complete adaptive transcript until a precise failure occurs. The lemma does not assert that the two executions have their original distributions *conditioned on* avoiding failure. Conditioning changes distributions and must be analysed separately.

4.8 Uniform maps, rejection and reduction bias

A bijection carries the uniform distribution to uniform. More generally, a function with exactly c preimages for every output carries uniform input to uniform output: the probability of each output is c divided by the input-set size. These facts need no computational assumption.

To sample uniformly from $\{0, \dots, m-1\}$, choose k with $2^k \geq m$, sample k independent bits, and reject values at least m . Each accepted value had the same probability before conditioning, so the result is exactly uniform. The expected number of trials is $2^k/m$. If instead one reduces the k -bit integer modulo m , write $2^k = qm + s$, $0 \leq s < m$. The first s residues have $q+1$ preimages, and the others have q . Direct substitution in the distance formula gives

$$\text{SD}(U_k \bmod m, U_m) = \frac{s(m-s)}{m2^k} \leq \frac{m}{4 \cdot 2^k}. \quad (5)$$

The inequality uses $s(m-s) \leq m^2/4$, obtained by completing the square. A long reduction input makes the bias small; it does not make it identically zero.

Example 4.5 (Visible modular bias). Reducing a uniform five-bit integer modulo seven produces preimage counts $(5, 5, 5, 5, 4, 4, 4)$. Its distance from uniform modulo seven is $3/56$. The checker counts the preimages and evaluates the distance using integers, so this example does not rely on floating-point rounding.

4.9 Algorithms, costs and security parameters

An algorithm is a finite description of operations on finite inputs, with a specified result or failure behaviour. A deterministic algorithm has no random choices; a randomised one also reads random bits. Its input length counts bits, not the numerical value of an encoded integer. Testing every candidate below a k -bit modulus takes on the order of 2^k iterations, even though the modulus occupies only k bits. Counting a field multiplication as one operation is an algebraic cost model; a bit-cost claim must additionally account for the size and implementation of the multiplication.

For nonnegative functions on positive integers, $f = O(g)$ means there are constants C, n_0 with $f(n) \leq Cg(n)$ for $n \geq n_0$. The notation $\Omega(g)$ reverses the bound, and $\Theta(g)$ asserts both. Logarithms $\log_2 n$ measure binary length up to rounding. Polynomial time means a bound Cn^d for fixed constants C, d , not constants chosen separately for each input. A probabilistic polynomial-time algorithm, abbreviated PPT, uses random bits and obeys such a time bound on every execution. Expected polynomial time is a different condition and must be named when used.

A security parameter λ indexes a family of instances and algorithms. A nonnegative function $\epsilon(\lambda)$ is negligible if, for every positive integer d , eventually $\epsilon(\lambda) < \lambda^{-d}$. A fixed inverse polynomial is not negligible. Exponential decay such as $2^{-\lambda}$ is negligible: for a fixed d , the ratio of consecutive terms of $\lambda^d/2^\lambda$ tends to one half, so eventually the sequence decreases geometrically to zero. Finite sums of

negligible functions, and their products by fixed polynomials, are negligible by choosing a stronger exponent in the definition.

An asymptotic claim does not supply a concrete attack cost for one fixed curve. A concrete statement records running time, query counts and success probability at the chosen parameters. Likewise, expected time bounded by T can be truncated after T/η steps with abort probability at most η , by Markov; truncation is a change to the algorithm whose failure probability belongs in the final bound.

5 Linear algebra and uniform masks

Linear algebra studies operations that preserve sums and multiplication by field elements. It supplies two different tools: efficient identities for rearranging long computations, and exact counting statements about which observations remain hidden by random masks.

5.1 Vectors, subspaces and bases

For a field $\mathbb{F}$, a vector in $\mathbb{F}^n$ is an ordered tuple $\mathbf{v} = (v_0, \dots, v_{n-1})$. Addition and scalar multiplication operate componentwise. A subspace $V \subseteq \mathbb{F}^n$ contains zero and is closed under those operations. A linear combination of vectors $\mathbf{v}_i$ is a sum $\sum_i a_i \mathbf{v}_i$. Their span is the subspace of all their linear combinations. The vectors are linearly independent if $\sum_i a_i \mathbf{v}_i = 0$ forces every coefficient to be zero. A basis is an independent spanning list.

The standard coordinate vectors $\mathbf{e}_i$, with a single one in position i , form a basis of $\mathbb{F}^n$. Relative to any basis, the coordinates of a vector are unique: subtracting two expressions would give a nonzero dependence. If a list is dependent, one vector whose coefficient in a dependence is nonzero can be expressed using the others and removed without changing the span. Removing such vectors from a finite spanning list constructs a basis.

Lemma 5.1 (Basis exchange). *If an independent list has k vectors and a spanning list has m vectors in the same space, then $k \leq m$. All bases of a space with a finite basis consequently have the same length.*

Proof. Express the first independent vector using the spanning list. Some coefficient is nonzero, so solve for that spanning vector and replace it by the independent vector, preserving the span. Repeat with the next independent vector. Its expression must use at least one not-yet-replaced vector with a nonzero coefficient; otherwise it would lie in the span of earlier independent vectors. Each step consumes a different member of the original spanning list, so there can be no more than m steps. Apply this in both directions to two bases. $\square$

The common basis length is the dimension. An independent list can be extended to a basis by adding vectors from a known spanning list when they increase the span. Thus every subspace of $\mathbb{F}^n$ has a basis of at most n vectors: start with an empty list and add a vector outside its span until none remains. More than n additions would contradict Lemma 5.1 applied in $\mathbb{F}^n$.

5.2 Linear maps, matrices and elimination

A linear map $L : V \rightarrow W$ satisfies $L(a\mathbf{v} + b\mathbf{w}) = aL(\mathbf{v}) + bL(\mathbf{w})$. It is determined by its values on a basis. In coordinate spaces these values become the columns of a matrix A ; the output coordinates are

$$(A\mathbf{x})_i = \sum_j A_{ij}x_j.$$

Matrix multiplication represents composition: $(AB)_{ij} = \sum_k A_{ik}B_{kj}$. Interchanging finite sums verifies $(AB)\mathbf{x} = A(B\mathbf{x})$. The identity matrix has ones on its diagonal and zeros elsewhere.

The kernel $\ker L$ is the subspace mapped to zero. The image is a subspace of the codomain; its dimension is the rank. A square matrix is invertible when its linear map is bijective. Its inverse represents the inverse map, so multiplication in either order gives the identity.

One can solve $A\mathbf{x} = \mathbf{b}$ constructively by row operations on the augmented matrix $(A \mid \mathbf{b})$. Swap two rows, multiply a row by a nonzero field element, or add a multiple of one row to another. Each operation has an inverse of the same form and preserves the set of solutions. Working from left to right, choose a nonzero entry as a pivot, move its row into position, scale it to one, and eliminate entries below it. A row with zero coefficients and a nonzero right side proves inconsistency. Otherwise, choose arbitrary values for the nonpivot variables and recover the pivot variables by back substitution. Eliminating above pivots as well gives an explicit description without back substitution.

Theorem 5.2 (Rank and nullity). *For a linear map $L : V \rightarrow W$ on a finite-dimensional domain,*

$$\dim V = \dim \ker L + \dim \operatorname{im} L.$$

Proof. Take a basis $\mathbf{k}_1, \dots, \mathbf{k}_s$ of the kernel and extend it to a basis of V by $\mathbf{v}_1, \dots, \mathbf{v}_t$. The images of these additional vectors span the image because the kernel basis maps to zero. They are independent: a combination with zero image would lie in the kernel and therefore express a dependence in the extended basis. Their number is the rank, giving the equality. $\square$

In particular, a square matrix is invertible exactly when its kernel is zero. Elimination tests this and constructs the inverse by applying the row operations to $(A \mid I)$. For an n -dimensional domain over a field with q elements, a rank- r linear map has a kernel with q^{n-r} elements, since each basis coordinate has q choices. Every nonempty fibre $L^{-1}(\mathbf{y})$ is a translate of that kernel: if $L(\mathbf{x}_0) = \mathbf{y}$, then $L(\mathbf{x}) = \mathbf{y}$ precisely when $\mathbf{x} - \mathbf{x}_0 \in \ker L$.

5.3 Inner products and point-valued linear maps

The field inner product is $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_i a_i b_i$. It is bilinear, meaning linear in either argument with the other fixed. It is also nondegenerate: if its value is zero for every $\mathbf{b}$, test $\mathbf{b} = \mathbf{e}_i$ to obtain $a_i = 0$ for every i . Unlike a sum of squares over the real numbers, this field-valued expression need not be nonzero on a nonzero vector paired with itself. For instance, a nonzero vector can have $\langle \mathbf{a}, \mathbf{a} \rangle = 0$ over a finite field. No argument may infer that a sum of field squares vanishes only when every term vanishes.

The mixed inner product $\langle \mathbf{a}, \mathbf{G} \rangle$ of §3 has a point as output. The same distribution laws permit splitting it into halves or changing coordinates. For an invertible field matrix M , transposition exchanges row and column indices, and

$$\langle M\mathbf{a}, M^{-T}\mathbf{b} \rangle = \langle \mathbf{a}, \mathbf{b} \rangle, \quad M^{-T} = (M^{-1})^T.$$

Expanding the products and using $M^T M^{-T} = I$ proves this identity. Point-valued versions are proved by the same finite sum expansion, using the scalar laws in the group. An implementation may use such identities to combine many checks, but the identities alone say nothing about which coefficients an adversary knows.

5.4 What a random linear mask hides

An affine map is a linear map followed by a translation: $\mathbf{x} \mapsto \mathbf{c} + L(\mathbf{x})$. A translate $\mathbf{c} + V = \{\mathbf{c} + \mathbf{v} : \mathbf{v} \in V\}$ of a subspace is an affine set. It need not contain zero and need not be a subspace. These definitions describe the exact output sets whose probabilities we now count.

Theorem 5.3 (Uniform image of an affine map). *Let R be uniform on $\mathbb{F}_q^n$, let $L : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^k$ have rank r , and fix $\mathbf{c} \in \mathbb{F}_q^k$. The vector $\mathbf{c} + L(R)$ is uniform on the affine set $\mathbf{c} + \text{im } L$, which has q^r elements. It is uniform on the entire codomain exactly when $r = k$.*

Proof. Every output in the image has q^{n-r} preimages by the fibre count above. Its probability is $q^{n-r}/q^n = q^{-r}$. Translation by $\mathbf{c}$ is a bijection. $\square$

If a secret random variable W is independent of R and L is surjective, then $C(W) + L(R)$ has the same uniform distribution for every possible value of W . Hence the output is independent of the secret. If the rank is smaller, the affine support may depend on the secret and reveal information even though every individual coordinate is uniform.

Example 5.4 (Marginal hiding is not joint hiding). Over $\mathbb{F}_5$, let u, v be independent uniform values. The map $(u, v) \mapsto (u + v, u + 2v)$ is bijective: subtracting the coordinates recovers v , then u . Adding any secret pair therefore leaves the output uniform on all 25 pairs. In contrast, $(u, v) \mapsto (u + v, 2u + 2v)$ has only five outputs. After adding a secret pair (c_1, c_2) , the observations satisfy $y_2 - 2y_1 = c_2 - 2c_1$. Each observation separately is uniform, yet together they reveal this secret linear combination. The checker enumerates both distributions exactly.

Previously disclosed linear information consumes masking dimensions. This can be quantified without an independence guess.

Lemma 5.5 (Masks after linear observations). *Let R be uniform on $\mathbb{F}_q^n$. Condition on $BR = \mathbf{b}$, for a value in the image of B . Then AR is uniform on a translate of $A(\ker B)$, and*

$$\dim A(\ker B) = \text{rank} \begin{pmatrix} B \\ A \end{pmatrix} - \text{rank } B.$$

In particular, k output coordinates of AR remain jointly uniform exactly when the displayed rank difference equals k .

Proof. Conditioning a uniform input on a fibre makes it uniform on that fibre, namely $\mathbf{r}_0 + \ker B$. Apply Theorem 5.3 to the restriction of A to $\ker B$. Rank-nullity identifies its rank as $\dim \ker B - \dim(\ker B \cap \ker A)$. Applying rank-nullity again to B and the stacked map proves the formula. $\square$

This lemma has fixed linear maps. If challenges or maps depend adaptively on previous observations, one must justify the conditional distribution at each step, including which fresh mask coordinates remain. Merely counting random coefficients in the original witness does not prove that a complete adaptive transcript is hidden.

6 Polynomials and evaluation

A polynomial turns a list of coefficients into a rule for obtaining field values. Its crucial property is that a nonzero polynomial of small degree cannot vanish at many different points. That fact supports both reconstruction from evaluations and probabilistic equality checks.

6.1 Formal polynomials and division

A polynomial over $\mathbb{F}$ is a finite formal sum $f(X) = \sum_{i=0}^d a_i X^i$. The symbol X is an indeterminate, not a sampled field element. Equality of polynomials means equality of all coefficients after appending zero coefficients as necessary. Addition is coefficientwise and multiplication is convolution:

$$(fg)_k = \sum_{i+j=k} a_i b_j.$$

Associativity, commutativity and distributivity follow by rearranging finite sums. The set of polynomials is written $\mathbb{F}[X]$. Evaluation substitutes $x \in \mathbb{F}$ for X , giving $f(x) = \sum_i a_i x^i$. Evaluation preserves addition and multiplication, again by expansion.

For nonzero f , its degree is the highest index with nonzero coefficient. The zero polynomial has degree $-\infty$, a convention that makes inequalities about upper degree bounds include zero. We never use an expression such as a negative number of roots for this special degree. For nonzero polynomials, multiplying the leading coefficients gives a nonzero leading coefficient, hence $\deg(fg) = \deg f + \deg g$. Also $\deg(f + g) \leq \max(\deg f, \deg g)$, with possible strict inequality when leading terms cancel.

Theorem 6.1 (Polynomial division). *For $f, g \in \mathbb{F}[X]$ with $g \neq 0$, unique polynomials q, r satisfy $f = qg + r$ and either $r = 0$ or $\deg r < \deg g$.*

Proof. If the current remainder has degree at least $\deg g$, subtract the unique scalar multiple of a power of X times g that cancels its leading term. This uses the inverse of the leading coefficient of g . Its degree strictly decreases, so the process terminates, and the subtracted multiples give q . For uniqueness, two outputs would give $(q - q')g = r' - r$. A nonzero left side has degree at least $\deg g$; a nonzero right side has smaller degree. They can therefore only both be zero. $\square$

Evaluation can be performed by Horner's rule. Starting from the highest coefficient, repeatedly multiply the accumulated value by x and add the next coefficient. Expanding the nested expression proves it equals $f(x)$, using d multiplications and d additions for a degree- d polynomial. Polynomial division, likewise, is an explicit finite algorithm, not a claim that a quotient happens to exist.

6.2 Roots, multiplicity and equality tests

Theorem 6.2 (Factor and root theorems). *For $a \in \mathbb{F}$, the equality $f(a) = 0$ holds exactly when $X - a$ divides f . A nonzero degree- d polynomial has at most d distinct roots in $\mathbb{F}$.*

Proof. Division by $X - a$ gives a constant remainder; evaluating at a identifies it with $f(a)$. For the root bound, a nonzero constant has no roots. If a positive-degree polynomial has a root a , write $f = (X - a)g$. Every other root is a root of g , since its $X - a$ factor is nonzero there. Induction on the degree proves the bound. $\square$

A root has multiplicity m if $(X - a)^m$ divides f but $(X - a)^{m+1}$ does not. We use this definition only for nonzero polynomials. Removing factors successively shows that the sum of multiplicities of distinct roots is at most the degree. The formal derivative is $f'(X) = \sum_{i \geq 1} i a_i X^{i-1}$, where the integer coefficient is interpreted in the field. Direct multiplication proves the product rule $(fg)' = f'g + fg'$. Thus a root of multiplicity at least two is also a root of f' . Conversely, if $f = (X - a)g$, then $f'(a) = g(a)$; if this vanishes, $X - a$ divides g , giving multiplicity at least two. These statements use a formal operation and require no calculus.

Over $\mathbb{F}_p$, the nonzero polynomial $X^p - X$ evaluates to zero at every field element, by Fermat and the separate zero case. Therefore polynomials and the functions they induce on a finite field are not the same objects. Degree restrictions are essential to uniqueness arguments.

Corollary 6.3 (A random evaluation test). *Fix different polynomials f, g of degree at most d , and then sample x uniformly from a nonempty finite set $S \subseteq \mathbb{F}$. The probability that $f(x) = g(x)$ is at most $\min(1, d/|S|)$.*

Proof. Apply the root bound to the nonzero difference. $\square$

The word “then” is a hypothesis. If the purported polynomial can be chosen after observing x , the adversary can choose $X - x$, which passes at that point with certainty. Any use of the corollary in a

protocol must explain how its polynomial is fixed before the sampled challenge, or invoke a separate argument that supplies that property.

6.3 Interpolation and vanishing polynomials

Let $H = \{x_0, \dots, x_{n-1}\}$ contain n distinct field elements, with $n \geq 1$. Define

$$Z_H(X) = \prod_{i=0}^{n-1} (X - x_i), \quad L_i(X) = \prod_{j \neq i} \frac{X - x_j}{x_i - x_j}.$$

All denominators are nonzero. The polynomials L_i have degree less than n and satisfy $L_i(x_j) = 1$ when $i = j$, zero otherwise.

Theorem 6.4 (Interpolation). *For any $y_0, \dots, y_{n-1} \in \mathbb{F}$, exactly one polynomial of degree less than n takes those values on the corresponding points. It is $f(X) = \sum_i y_i L_i(X)$.*

Proof. The evaluation property of the L_i proves existence. The difference of two candidates would have n distinct roots and degree less than n , so the root bound forces it to be zero. $\square$

The space of degree-less-than- n polynomials has coefficient basis $1, X, \dots, X^{n-1}$. Interpolation proves it also has basis $L_0, \dots, L_{n-1}$: these polynomials span by the formula and are independent by evaluation at each x_i . Evaluation is therefore an invertible linear change of coordinates. Its matrix is $V_{ij} = x_i^j$, often called a Vandermonde matrix; its invertibility follows from the interpolation argument just given.

Proposition 6.5 (Vanishing on a domain). *A polynomial f is zero at every point of H exactly when $Z_H \mid f$. Moreover,*

$$L_i(X) = \frac{Z_H(X)}{(X - x_i)Z'_H(x_i)}.$$

The quotient by $X - x_i$ in this expression is a polynomial.

Proof. Remove the factor $X - x_0$. At any other root, this factor is nonzero, so the remaining polynomial still vanishes there. Repeat to remove all factors. The reverse implication follows by evaluation. The product rule gives $Z'_H(x_i) = \prod_{j \neq i} (x_i - x_j) \neq 0$, proving the formula. $\square$

Polynomial division now has an evaluation interpretation. The unique remainder of f modulo Z_H is its degree-less-than- n interpolant on H , because subtraction of a multiple of Z_H does not change those values. This is also the precise source of aliasing: values on H determine a remainder, not an unrestricted high-degree polynomial.

6.4 Random polynomial masks and threshold observations

Choose the d coefficients of a polynomial of degree less than d independently and uniformly in $\mathbb{F}_q$. At any $k \leq d$ distinct points its evaluations are jointly uniform in $\mathbb{F}_q^k$. Indeed, restricting the evaluation map to the first k coefficients already gives an invertible interpolation matrix, so the full map is surjective. Apply Theorem 5.3.

Suppose instead that a polynomial must vanish on a fixed set H . Write it as $Z_H(X)r(X)$, where r has d uniform coefficients. At $k \leq d$ distinct points outside H , its evaluations remain jointly uniform: multiplication by the nonzero values $Z_H(x_i)$ is an invertible diagonal map. At a point inside H , the output is necessarily zero. Repeated observations at the same point are not additional independent masks; their evaluation rows coincide. Linear observations already made must be accounted for using Lemma 5.5.

Lemma 6.6 (A hidden constant coefficient). *Fix a value $s \in \mathbb{F}_q$ and let $f(X) = s + \sum_{i=1}^{t-1} a_i X^i$, with independent uniform coefficients. For any $k < t$ distinct nonzero evaluation points, the k evaluations have a uniform joint distribution independent of s . Conversely, evaluations at t distinct points determine the complete polynomial and hence s .*

Proof. For the first claim, the columns corresponding to $X, \dots, X^k$ give a matrix equal to the invertible Vandermonde matrix for $1, \dots, X^{k-1}$, with each row multiplied by its nonzero evaluation point. It has full row rank. The affine-mask theorem removes dependence on the constant shift. The second claim is interpolation. $\square$

When reconstruction needs only the constant coefficient, interpolation gives the explicit formula

$$f(0) = \sum_i \lambda_i f(x_i), \quad \lambda_i = \prod_{j \neq i} \frac{-x_j}{x_i - x_j}.$$

The index set in these products is the exact set of observations being used. Changing that set changes the coefficients. The nonzero-point condition in the hiding lemma is essential: an observation at zero reveals s immediately.

6.5 Several variables and random combinations

A polynomial in m variables is a finite sum of monomials $cX_1^{e_1} \dots X_m^{e_m}$. Its total degree is the largest sum $e_1 + \dots + e_m$ with nonzero coefficient. We may regard it as a polynomial in the last variable whose coefficients are polynomials in the others.

Theorem 6.7 (Schwartz–Zippel). *For a fixed nonzero polynomial of total degree at most d , and independent uniform $x_1, \dots, x_m$ in a finite nonempty set $S \subseteq \mathbb{F}$, the probability of a zero evaluation is at most $\min(1, d/|S|)$.*

Proof. The case $m = 1$ is the root bound. For the induction step, let k be the degree in X_m and $g \neq 0$ its leading coefficient polynomial. Its total degree is at most $d - k$, so the probability that it vanishes on the first $m - 1$ choices is at most $(d - k)/|S|$. Otherwise the remaining univariate polynomial is nonzero of degree k , and its conditional root probability is at most $k/|S|$, because x_m is still independent uniform. Add the two bounds and cap at one. $\square$

To combine fixed scalar equations $e_0 = 0, \dots, e_{k-1} = 0$, choose a fresh scalar η and test $\sum_i \eta^i e_i = 0$. If at least one equation is false, this is a nonzero polynomial of degree at most $k - 1$ in η . Independent uniform coefficients would instead give error exactly $1/q$ over $\mathbb{F}_q$, by solving for the coefficient of a nonzero e_i . The two methods have different bounds.

For polynomial equations, first apply the same reasoning to a coefficient that is nonzero in at least one equation; then account for the later random evaluation of the resulting polynomial. These are separate bad events. Random aggregation does not turn a single root-bound calculation into a security proof for the complete interactive protocol in which the equations arose.

7 Evaluation domains and fast transforms

Interpolation works on any distinct points, but a geometric sequence of points makes both evaluation and interpolation much faster. We construct those sequences and the transforms rather than treating the fast Fourier transform, abbreviated FFT, as an unexplained operation.

7.1 Roots of unity and their existence

An element $\omega \in \mathbb{F}^\times$ is a primitive n -th root of unity if its multiplicative order is exactly n . Its powers $1, \omega, \dots, \omega^{n-1}$ are then distinct, form a subgroup, and are roots of $X^n - 1$. The root bound proves that there are no further roots and hence $Z_H(X) = X^n - 1$ for this set H .

Theorem 7.1 (Cyclicity of a finite multiplicative subgroup). *Every finite subgroup of a field's multiplicative group is cyclic. In particular, $\mathbb{F}_q^\times$ has an element of order $q - 1$.*

Proof. Let M be the least common multiple of the orders of all elements. Factor M into its maximal prime-power factors. For each such factor ℓ^a , some element has order divisible by ℓ^a . Raising that element to the part of its order not divisible by ℓ gives an element of order ℓ^a . This follows directly from scalar reduction: for an element of order r , the order of its k -th power is $r/\gcd(r, k)$.

Two integers are coprime when their greatest common divisor is one. The product of commuting elements of coprime orders has order the product. If $(ab)^t = 1$, then $a^t = b^{-t}$ lies in the intersection of the two cyclic subgroups. Its order divides both coprime orders by Lagrange, so it is the identity; each order therefore divides t . Multiplying the elements constructed for all prime powers yields an element of order M .

Every member of the original subgroup is a root of $X^M - 1$, so its size is at most M by the root bound. The element just constructed already contributes M distinct powers. Thus the subgroup consists exactly of those powers. $\square$

There is a primitive n -th root in $\mathbb{F}_q$ exactly when $n \mid (q - 1)$: necessity follows from Lagrange, and for sufficiency raise a multiplicative generator to $(q - 1)/n$. The largest s for which $2^s \mid (q - 1)$ is the two-adicity of the field. An advertised root of order 2^s can be checked without finding a full generator: verify $\omega^{2^s} = 1$ and $\omega^{2^{s-1}} \neq 1$ when $s \geq 1$. When $s = 0$, the unique element of order one is simply 1. For general n , verify $\omega^n = 1$ and $\omega^{n/\ell} \neq 1$ for each prime divisor ℓ of n . Every proper divisor of n divides some n/ℓ , which proves the test.

7.2 The transform and its inverse

The characteristic of a finite field is the least positive integer ℓ for which $\ell \cdot 1 = 0$. It exists because the additive group is finite. It is prime: a factorisation $\ell = ab$ with $1 < a, b < \ell$ would produce two nonzero elements $a \cdot 1, b \cdot 1$ with zero product. Its additive subgroup generated by one has order ℓ , so Lagrange shows $\ell \mid q$ for a field of size q .

Let $n \mid (q - 1)$ and let ω have order n . The field element represented by n is nonzero and invertible: the characteristic prime divides q , not $q - 1$. For an integer k , a geometric sum gives

$$\sum_{j=0}^{n-1} \omega^{jk} = \begin{cases} n, & n \mid k, \\ 0, & n \nmid k. \end{cases} \quad (6)$$

In the second case multiply the sum by $1 - \omega^k \neq 0$; the product is $1 - \omega^{nk} = 0$. This proves the identity.

For coefficients $a_0, \dots, a_{n-1}$, the transform is $y_j = \sum_i a_i \omega^{ij}$, evaluation at ω^j . Its inverse is

$$a_i = n^{-1} \sum_{j=0}^{n-1} y_j \omega^{-ij}. \quad (7)$$

Substitute the expression for y_j , interchange the two sums, and use (6). Only the term with the matching coefficient index remains. In particular, the Lagrange basis here is

$$L_j(X) = \frac{1}{n} \sum_{i=0}^{n-1} \omega^{-ij} X^i = \frac{\omega^j}{n} \frac{X^n - 1}{X - \omega^j}.$$

The second expression follows from the derivative $(X^n - 1)' = nX^{n-1}$ and Proposition 6.5. At $X = \omega^j$ it denotes the polynomial quotient, not a numerical division by zero.

7.3 The radix-two algorithm

For even n , separate the even and odd coefficients:

$$f(X) = f_{\text{even}}(X^2) + Xf_{\text{odd}}(X^2).$$

The root ω^2 has order $n/2$. Compute the two length- $n/2$ transforms to obtain values E_j, O_j at ω^{2j} . For $0 \leq j < n/2$, combine them as

$$y_j = E_j + \omega^j O_j, \quad y_{j+n/2} = E_j - \omega^j O_j.$$

The identity $\omega^{n/2} = -1$ follows because it has order two, and the only roots of $X^2 - 1$ in a field of odd characteristic are $1, -1$. These formulas compute the two required evaluations. When $n = 2^k$, repeat the split down to length one, where a constant evaluates to itself.

At each of k levels the total number of field operations is bounded by a fixed multiple of n ; the result costs $O(n \log n)$, instead of evaluating one degree- $n - 1$ polynomial at n points by separate quadratic total work. The inverse uses the same algorithm with root ω^{-1} , followed by multiplication of each result by n^{-1} . An in-place implementation may reorder entries; the output contract remains the explicit indexing in (7).

Example 7.2 (A complete transform). In $\mathbb{F}_{97}$, the element 22 has order four, with powers (1, 22, 96, 75). The polynomial $3 + 5X + 7X^2 + 11X^3$ evaluates on this domain to (26, 58, 91, 31). Applying the inverse transform recovers (3, 5, 7, 11). Both directions are checked independently in the accompanying script.

7.4 Products, rotations and cosets

To multiply polynomials of degrees at most a, b , choose an available transform length $N > a + b$, pad both coefficient lists with zeros, evaluate them, multiply corresponding values, and interpolate. The true product has degree less than N , so interpolation uniquely recovers it. If $N \leq a + b$, the same procedure instead returns the remainder modulo $X^N - 1$. Calling it the full product would be incorrect.

For a polynomial f , the rotation $f(\omega X)$ evaluates on H to the cyclic shift $f(\omega^{j+1})$. Its coefficient of X^i is $a_i \omega^i$. The wrap at the last row is an algebraic property of the domain, not a decision about which circuit rows should participate in a constraint.

A multiplicative coset is $gH = \{g\omega^j\}$, with $g \neq 0$. Evaluating f on it is the ordinary transform of coefficients $a_i g^i$. Interpolation reverses this scaling. Its vanishing polynomial is $X^n - g^n$. If $g \notin H$, then $g^n \neq 1$, and the coset is disjoint from H .

Suppose a quotient $t = f/(X^n - 1)$ exists and has degree less than N , where $n \mid N$ and a size- N subgroup H_N is available. Choose $g \notin H_N$. Every point of gH_N lies outside H , because otherwise g would belong to H_N . Thus

$$t(x) = \frac{f(x)}{x^n - 1} \quad (x \in gH_N)$$

has no zero denominator and can be interpolated on that coset. The bound needed for this interpolation is on $\deg t$, not on $\deg f$. To obtain the numerator values one may evaluate its constituent low-degree polynomials on the coset and perform the specified pointwise expression; one need not interpolate the high-degree numerator itself in a space that is too small for it.

There is a converse trap. Pointwise division on a coset followed by interpolation always returns some polynomial, even when f is not divisible by $X^n - 1$. That does not prove the polynomial identity $f = (X^n - 1)t$. Divisibility must be established separately or tested with the appropriate degree and challenge conditions.

7.5 Chunks and recombination

Every polynomial of degree less than kn decomposes uniquely as

$$t(X) = \sum_{j=0}^{k-1} X^{jn} t_j(X), \quad \deg t_j < n,$$

by grouping consecutive blocks of n coefficients. At a point z , its value is $\sum_j z^{jn} t_j(z)$. This is a deterministic identity, with no randomness or probability loss. It allows a large quotient to be represented by several polynomials with a common smaller degree bound. A later protocol may commit to the chunks separately; such a commitment does not alter the recombination formula or the need for the degree bounds.

8 Pasta curves and their encodings

We now construct the point arithmetic used throughout the cryptographic volumes. Two distinctions remain visible: coordinates are not scalars, and an algebraic formula valid away from exceptional inputs is not a total group operation. The finite-field algorithms already developed make every ordinary arithmetic step explicit.

8.1 The curve and the complete affine operation

An affine coordinate pair means two ordinary field coordinates (x, y) , without identifying different pairs by a scaling rule. We first specify the curve and its operation in those coordinates, treating its extra identity as a separate object.

For a field $\mathbb{F}_p$ with prime $p > 3$, choose $A, B \in \mathbb{F}_p$ with $4A^3 + 27B^2 \neq 0$. Consider the finite coordinate pairs satisfying

$$y^2 = x^3 + Ax + B, \tag{8}$$

and add a separate point $\mathcal{O}$, called the point at infinity. This equation is called short-Weierstrass form. The nonvanishing condition is a hypothesis of the group theorem below. To see its algebraic meaning, a repeated root u of $X^3 + AX + B$ would satisfy $3u^2 + A = 0$ and $u^3 + Au + B = 0$, hence $A = -3u^2, B = 2u^3$, giving $4A^3 + 27B^2 = 0$.

Negate a finite point by $-(x, y) = (x, -y)$, and set $-\mathcal{O} = \mathcal{O}$. Define addition with the identity by $P + \mathcal{O} = \mathcal{O} + P = P$. For finite points $P = (x_1, y_1), Q = (x_2, y_2)$, use the following cases:

1. If $x_1 = x_2$ and $y_1 = -y_2$, return $\mathcal{O}$. This includes doubling a point with $y_1 = 0$.
2. Otherwise, if $P = Q$, set $\lambda = (3x_1^2 + A)/(2y_1)$.
3. Otherwise, set $\lambda = (y_2 - y_1)/(x_2 - x_1)$.

In the last two cases return

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1. \tag{9}$$

The cases cover all inputs. With equal x -coordinates, the curve equation gives $y_1^2 = y_2^2$; factoring their difference shows the points are equal or opposite. The excluded cases ensure every displayed denominator is nonzero.

These formulas come from intersecting a line with the cubic. Substitute the line $y = \lambda(x - x_1) + y_1$ into (8). After moving all terms to one side, the coefficient of x^2 is $-\lambda^2$. Two roots are x_1, x_2 , so the third is $\lambda^2 - x_1 - x_2$, by expanding the product of the three linear factors. Reflection in the x -axis gives y_3 . For equal inputs, the slope is chosen so that x_1 is a repeated root of the substituted polynomial; the formal derivative condition gives $2y_1\lambda = 3x_1^2 + A$. This proves that the returned finite point is on the curve and explains both ordinary addition and doubling without division by a vanishing denominator.

Theorem 8.1 (Imported elliptic-curve group theorem). *For $p > 3$ and $4A^3 + 27B^2 \neq 0$, the points of (8), together with $\mathcal{O}$, form an abelian group under the complete operation just defined.*

Closure was established above; the identity, inverses and symmetry of the construction are explicit. Associativity, including all exceptional cases, is the imported part of this theorem. The line-intersection picture does not prove it, and checking a few triples numerically does not replace it. We use the standard short-Weierstrass group theorem also assumed by the protocol specification’s section “Pallas and Vesta”. This is an explicit mathematical boundary of this volume, not a computational hardness assumption and not a claim that the preceding coefficient calculation proves the entire theorem.

8.2 Incomplete addition and circuit equations

The incomplete operation $P \boxplus Q$ is defined only for finite points with different x -coordinates. It uses the last slope formula and (9); on any other input it returns $\perp$. Thus it agrees with complete addition wherever defined, but it rejects $P = Q$, $P = -Q$, and any identity input. Equality of two finite x -coordinates is equivalent to equality up to negation, as proved above.

One can replace the division in its slope by the equation

$$(x_2 - x_1)\lambda = y_2 - y_1. \quad (10)$$

With $x_2 \neq x_1$, this equation uniquely determines the correct slope. With $P = Q$, it becomes $0 = 0$ and leaves λ unconstrained. Consequently, polynomial equations obtained by clearing denominators do not automatically enforce the domain of an incomplete operation. A circuit may need a nonzero check, or its surrounding security argument may need to account explicitly for exceptional witnesses allowed by the implemented relation. A description of a partial function and a description of its circuit constraints must not silently be identified.

A nonzero check can introduce z and enforce $xz = 1$: an inverse exists precisely when $x \neq 0$. A conditional nonzero check with a Boolean flag b , meaning an element constrained to zero or one, instead needs a correctly specified conditional relation, since $xz = b$ alone does not force $x = 0$ when $b = 0$. Such details are statements about equations, not minor implementation conventions.

8.3 Projective representation and scalar evaluation

Affine formulas require inversions. A projective coordinate system introduces a shared denominator and identifies scaled representatives. In ordinary projective coordinates, a point is an equivalence class of nonzero triples (X, Y, Z) , where (X, Y, Z) and (uX, uY, uZ) represent the same point for every nonzero u . For $Z \neq 0$, the corresponding affine coordinates are $(X/Z, Y/Z)$. The homogeneous curve equation, meaning that every term has the same total degree, is

$$Y^2Z = X^3 + AXZ^2 + BZ^3.$$

It is unchanged by the scaling because every term acquires a factor u^3 . With $Z = 0$, the equation forces $X = 0$, and the nonzero-triple condition leaves the single equivalence class of $(0, 1, 0)$. This constructs the extra point at infinity in the same coordinate language as the finite points.

Jacobian coordinates use a different scaling rule, chosen to reduce arithmetic costs. A finite point is represented by (X, Y, Z) , with $Z \neq 0$, meaning $x = X/Z^2$, $y = Y/Z^3$. Its equation becomes

$$Y^2 = X^3 + AXZ^4 + BZ^6.$$

Scaling the triple to (u^2X, u^3Y, uZ) , with $u \neq 0$, does not change the represented point. This is a weighted scaling: the powers of u are two, three and one rather than all one. For the algorithms

here the identity is a separate tagged value. An implementation may instead reserve particular zero- Z triples for it; that internal convention is not an ordinary affine point and must be handled before division.

To see how denominator removal works, consider two finite triples. Set

$$U_1 = X_1 Z_2^2, \quad U_2 = X_2 Z_1^2, \quad S_1 = Y_1 Z_2^3, \quad S_2 = Y_2 Z_1^3,$$

and $H = U_2 - U_1$, $R = S_2 - S_1$. When $H \neq 0$, return

$$\begin{aligned} X_3 &= R^2 - H^3 - 2U_1 H^2, \\ Y_3 &= R(U_1 H^2 - X_3) - S_1 H^3, \\ Z_3 &= H Z_1 Z_2. \end{aligned}$$

Dividing these expressions by Z_3^2, Z_3^3 respectively recovers (9), because $x_2 - x_1 = H/(Z_1^2 Z_2^2)$ and $y_2 - y_1 = R/(Z_1^3 Z_2^3)$. This is an explicit inversion-free ordinary-addition formula, not a complete formula for $H = 0$. Doubling, opposites and identities require their own cases or a separately verified complete projective formula.

For correctness, one may always convert to affine and use the complete operation already specified. For performance, an implementation retains projective coordinates through a sequence of additions and normalises once at the end. Batch normalisation can use the simultaneous inversion algorithm in §2. No choice of coordinates alters the distinction between the coordinate field and the scalar modulus.

8.4 Quadratic residues and square-root extraction

A nonzero element $a \in \mathbb{F}_p$ is a square if $a = y^2$ for some y . A nonzero square has exactly two square roots $y, -y$, because $X^2 - a$ has at most two roots and the two are distinct when p is odd. The squaring map on $\mathbb{F}_p^\times$ is therefore two-to-one onto a set of size $(p-1)/2$. By cyclicity, writing $a = g^k$ for a generator g , the squares are exactly the even powers. The element $g^{(p-1)/2}$ is -1 , so

$$a^{(p-1)/2} = \begin{cases} 1, & a \text{ is a nonzero square,} \\ -1, & a \text{ is a nonsquare.} \end{cases} \quad (11)$$

This is Euler's criterion, including an efficient test by exponentiation. The zero input has the unique square root zero and is handled separately.

Here is a constructive algorithm valid for every odd prime. Write $p-1 = 2^s d$, with d odd, and choose a nonsquare z , verified with (11). For a nonzero input a , first reject unless Euler's test gives one. Initialise

$$x = a^{(d+1)/2}, \quad t = a^d, \quad c = z^d, \quad m = s.$$

While $t \neq 1$, find the least $i \geq 1$ with $t^{2^i} = 1$, set $b = c^{2^{m-i-1}}$, and update

$$x \leftarrow xb, \quad t \leftarrow tb^2, \quad c \leftarrow b^2, \quad m \leftarrow i.$$

Return x . This is the Tonelli–Shanks algorithm.

Proposition 8.2 (Square-root algorithm correctness). *For the accepted inputs above, the loop is defined, terminates and returns a square root of a . It uses $O(\log p + s^2)$ field multiplications after a verified nonsquare has been supplied. A search for that nonsquare is additional work; independent uniform trials need two trials on average, with an $O(\log p)$ Euler test per trial. This is an expected bound, not a fixed worst-case bound for deterministic sequential search.*

Proof. Maintain $x^2 = at$, $\text{ord}(c) = 2^m$, and $\text{ord}(t) \mid 2^{m-1}$. Initially the first identity is a direct power calculation. Euler’s criterion gives $c^{2^{s-1}} = -1$, so c has order 2^s , and $t^{2^{s-1}} = 1$ since a is a square. Thus all invariants hold.

When $t \neq 1$, its order is 2^i with $1 \leq i < m$, so the exponent defining b is nonnegative. The element b has order 2^{i+1} , and b^2 has order 2^i . Both t and b^2 , raised to 2^{i-1} , equal the unique order-two element -1 . Their product consequently has order dividing 2^{i-1} . The updated variables satisfy all three invariants with the strictly smaller value $m = i$. Termination follows, and the first invariant gives $x^2 = a$ when $t = 1$. Initial exponentiations cost $O(\log p)$; each of at most s iterations uses at most $O(s)$ squarings or multiplications. $\square$

A nonsquare can be found by testing candidates, or by independent uniform sampling from $\mathbb{F}_p^\times$, which succeeds with probability one half per trial. The variable-time algorithm above specifies an arithmetic result; an implementation handling secrets must separately prevent execution time or memory accesses from revealing sensitive choices. A final check $x^2 = a$ is a useful implementation assertion but does not replace input validation.

8.5 Concrete parameters and the two field roles

Define

$$\begin{aligned} p_{\text{Pallas}} &= 2^{254} + 45560315531419706090280762371685220353, \\ p_{\text{Vesta}} &= 2^{254} + 45560315531506369815346746415080538113. \end{aligned}$$

The curve E_{Pallas} has equation $y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\text{Pallas}}}$; the curve E_{Vesta} has the same equation over $\mathbb{F}_{p_{\text{Vesta}}}$. The short-Weierstrass nonvanishing condition is satisfied because $27 \cdot 25 \neq 0$ in either field.

Assumption 8.3 (Imported, verifiable parameter facts). Both displayed integers are prime, and the complete point groups have orders

$$|E_{\text{Pallas}}(\mathbb{F}_{p_{\text{Pallas}}})| = p_{\text{Vesta}}, \quad |E_{\text{Vesta}}(\mathbb{F}_{p_{\text{Vesta}}})| = p_{\text{Pallas}}.$$

These are mathematical parameter facts imported from the Zcash protocol specification, “Pallas and Vesta”, not assumptions of computational intractability.

The specification is available at the primary protocol source. The inspected implementation is `pasta_curves` version 0.5.2, with moduli in `src/fields/fp.rs` and `src/fields/fq.rs`, and group/type definitions in `src/curves.rs`. These identify the rule set being explained; they do not by themselves establish the activation of any network upgrade.

Under these parameter facts, every nonidentity point generates its group and both cofactors are one. The Pallas base field, used for coordinates, is $\mathbb{F}_{p_{\text{Pallas}}}$, but its scalar field is $\mathbb{F}_{p_{\text{Vesta}}}$. Vesta reverses these roles. This is the Pasta cycle: the base field of each curve is the scalar field of the other. The cycle makes Pallas coordinate operations expressible directly in arithmetic over Vesta’s scalar field. It does not identify Pallas scalars with Pallas coordinates, and it does not make all operations in an algorithm using both curves become operations in a single field.

Both fields have two-adicity exactly 32: divide the corresponding $p - 1$ by 2^{32} , and the remaining integer is odd. The checker verifies this, as well as Euler’s criterion showing that -13 and 5 are nonsquares in both fields. The point $(-1, 2)$ satisfies $4 = (-1)^3 + 5$, so it is a nonidentity point in each group. The checker also verifies that multiplication by the published opposite-field order returns the identity. That check confirms a scalar-multiplication identity; it is not a primality certificate or a proof that there are no additional curve points. The imported order facts remain explicit.

8.6 Compressed points and coordinate extraction

Let p be the relevant base modulus. For a finite point $P = (x, y)$, write $\tilde{y}$ for the low bit of the canonical integer representative of y . The star encoding is

$$P^* = \text{LE}_{256}(\text{val}(x) + 2^{255}\tilde{y}), \quad \mathcal{O}^* = \text{LE}_{256}(0). \quad (12)$$

Here $\text{val}(x)$ denotes the canonical integer in $[0, p)$. Packing each consecutive eight bits into a byte gives the 32-byte representation. The top bit carries the sign selector; the lower 255 bits carry the x -coordinate.

Decoding performs the following exact checks:

1. Require exactly 32 bytes. Extract the top bit s , and interpret the remaining 255 bits as an integer u .
2. Reject if $u \geq p$. Otherwise set $x = [u]_p$.
3. If $u = 0$ and $s = 0$, return $\mathcal{O}$.
4. Compute a square root y of $x^3 + 5$. Reject if no root exists. Select y or $-y$ so that the canonical integer's low bit equals s , then return (x, y) .

This is the decoding behaviour in `pasta_curves 0.5.2, src/curves.rs`, implementing the specification's "Pallas and Vesta" encoding. In particular, reduction of a noncanonical $u \geq p$ would be incorrect.

There are no finite points with $y = 0$, because such a point would be a nonidentity element of order two in a group of odd prime order. The two square roots in step four are therefore distinct and have opposite parity, since their canonical representatives sum to the odd integer p . There are also no finite points with $x = 0$, because 5 is a nonsquare in these fields. Consequently, the identity encoding does not collide with a finite point, and the input with only the top bit set is invalid rather than another identity encoding. Every valid group point has exactly one valid byte representation.

Define coordinate extraction by $\text{Extract}(\mathcal{O}) = 0$ and $\text{Extract}(x, y) = \text{val}(x)$. Extending it to a partial operation preserves failure: $\text{Extract}(\perp) = \perp$, not zero. Equality of extracted coordinates for two finite points means the points are equal or opposite; it does not generally mean the full points are equal. For Pasta, extracted zero identifies the identity, because finite $x = 0$ points do not exist. These exact facts are required whenever a commitment is shortened to one coordinate.

8.7 A small curve and the limits of examples

Over $\mathbb{F}_{19}$, the equation $y^2 = x^3 + 5$, together with the identity, has 27 points. Direct enumeration of all 361 coordinate pairs establishes this count in the checker. The point $(0, 9)$ doubles to $(0, 10)$, its negative, and its third multiple is the identity. This illustrates the complete operation and a non-prime curve order. It does not use Pasta's compressed-identity convention, whose disjointness relied on the absence of finite $x = 0$ points.

No small example establishes a security level. Even the exact Pasta group-order facts would not prove that discrete logarithms are hard. They establish the algebraic domains on which a later security assumption can be stated. The imported group theorem and parameter facts are the only substantial unproved mathematical inputs to this curve construction; all encoding operations and ordinary arithmetic algorithms used here have been given explicitly.

9 Byte fields for AES

A byte-oriented algorithm may use a field that is not a prime field. For AES, bytes represent polynomials with binary coefficients, not integers modulo 256. This final construction supplies exactly that field and its invertible operations.

9.1 Binary polynomial arithmetic

Use $\mathbb{F}_2 = \{0, 1\}$, where addition and subtraction coincide with exclusive-or. Represent a byte a by

$$a(X) = a_0 + a_1X + \cdots + a_7X^7,$$

where a_i are the bits of its integer encoding. Define $M(X) = X^8 + X^4 + X^3 + X + 1$. Add polynomials coefficientwise and multiply them by convolution, then reduce the result modulo M using polynomial division. The result again has degree less than eight and so fits in one byte. Replacing either input by a congruent polynomial does not change the result, because multiples of M remain multiples under addition and multiplication. Associativity and distributivity therefore descend from $\mathbb{F}_2[X]$.

A polynomial of positive degree is irreducible if it is not a product of two polynomials of smaller positive degree. A reducible degree-eight polynomial has a factor of degree at most four: in any factorisation, at least one of the two degrees is at most four. Since the only nonzero leading coefficient in $\mathbb{F}_2$ is one, all candidate factors are monic, meaning that their leading coefficient is one. There are $2 + 4 + 8 + 16 = 30$ monic polynomials of degrees one through four. Dividing M by each is a complete finite irreducibility test. The accompanying checker performs all 30 divisions and verifies that none has zero remainder. The hexadecimal representation of the full modulus is 0x11b; the ninth bit records the X^8 term and is not itself an eight-bit field element.

Proposition 9.1 (The AES field). *The 256 degree-less-than-eight binary polynomials, with the operations above, form a field.*

Proof. We have established the arithmetic laws inherited from polynomials. To invert a nonzero polynomial a of degree less than eight, run Euclid's algorithm on a, M , using polynomial division. The last nonzero remainder is a common divisor and a polynomial combination $ua + vM$, by exactly the coefficient bookkeeping in the integer algorithm. A nonconstant common divisor would factor the irreducible polynomial M ; its degree cannot be eight because it also divides a . Thus the greatest common divisor is the constant one. Reducing the identity $ua + vM = 1$ modulo M gives the inverse $u \bmod M$. $\square$

The field contains a copy of $\mathbb{F}_2$: send zero and one to their constant-polynomial representatives. This map preserves one, addition and multiplication, and is injective. Its image is therefore a subfield, a subset that is itself a field under the inherited operations. We may identify that image with the original $\mathbb{F}_2$.

A field containing a specified copy of another field is a field extension of that smaller field. Thus the byte field is an extension of $\mathbb{F}_2$, not merely a larger set carrying the same names for some elements. Multiplication by a constant polynomial makes it a vector space over $\mathbb{F}_2$. Its eight representatives $1, X, \dots, X^7$ form a basis: every reduced polynomial is their linear combination, and a nonzero combination of degree less than eight cannot be a multiple of the degree-eight modulus. The dimension of this vector space is the degree of the field extension, here eight. Consequently it has exactly 2^8 elements, one for each choice of eight binary coefficients. This derives both the extension terminology and the size from the constructed operations; no unspecified extension arithmetic is being used.

We denote this field by $\mathbb{F}_{2^8}$. Its concrete byte representation depends on the chosen modulus and basis. A field isomorphism preserves its mathematical operations, as defined in §2, but need not preserve the integer labels of the bytes. Consequently another byte basis can give a different encoded multiplication table. The symbol 0x02 represents X , not the prime-field integer two, which is zero in characteristic two.

9.2 Constructive multiplication and inversion

Multiplication by X shifts the byte left by one bit. If the old highest bit was one, the shift introduced X^8 , which reduces to $X^4 + X^3 + X + 1$. Therefore the byte operation is

$$\text{xtime}(a) = ((a \ll 1) \bmod 256) \oplus \begin{cases} 0x1b, & a \geq 128, \\ 0, & a < 128. \end{cases}$$

To multiply a by b , initialise a zero accumulator and scan the eight bits of b from low to high. Exclusive-or the current value of a into the accumulator for each set bit, and apply xtime to a after each step. This is polynomial convolution followed by reduction, so it proves the operation implements the field product.

The nonzero multiplicative group has order 255. Lagrange gives $a^{255} = 1$, so $a^{-1} = a^{254}$ for nonzero a . Exponentiation using the byte multiplication just specified is a second complete inversion algorithm. A later substitution operation may define an extended inverse that maps zero to zero. That is a separate total function, not a claim that zero has a multiplicative inverse.

Example 9.2 (Byte arithmetic). The field product $0x57 \cdot 0x83$ is $0xc1$, and the inverse of $0x53$ is $0xca$. The checker evaluates both through the explicit polynomial arithmetic and checks $aa^{254} = 1$ for all 255 nonzero byte values.

An affine map on the eight bits means a binary linear transformation followed by exclusive-or with a fixed byte. To test its invertibility, form the eight-by-eight matrix of its effect on the coordinate bits and apply elimination over $\mathbb{F}_2$. A byte substitution formed by the extended inverse followed by an invertible affine map is a permutation: the extended inverse exchanges each nonzero element with its inverse and fixes zero; composition preserves bijectivity. This establishes the algebra needed to construct AES's substitution step, without identifying byte-field arithmetic with ordinary integer arithmetic or assuming a cipher's security from its invertibility.

Verification boundary

The numerical examples in this volume are produced and checked by `check_mathcrypto.py` alongside the rewrite sources. The checks include field inverses and reduction bias, rank-one and rank-two mask distributions, both directions of the four-point transform, a complete small-curve point count, published Pasta scalar identities, and exhaustive byte-field inverses and modulus irreducibility. They do not certify Pasta primality or point counts, prove the imported elliptic-curve associativity theorem, establish cryptographic hardness, or certify constant-time implementations. Those limits are part of the mathematical contract, not omissions to be filled by trusting a worked example.